MRI Instrumentation

First: What is MRI?

MRI = **Magnetic Resonance Imaging**

medical imaging technique used to visualize internal structures of the body.

Basic Principle of MRI

MRI is based on:

1- A strong magnetic field
2- Radiofrequency (RF) waves
3- Hydrogen nuclei

Why Hydrogen?

Because:

1- It is abundant in the human body
2- It has a strong magnetic moment

Steps of MRI Operation

① The patient is placed inside a strong magnet
② Hydrogen nuclei align with the magnetic field
③ Radio waves are applied

④ The nuclei absorb energy

⑤ The RF pulse is turned off

⑥ The nuclei return to their original state

⑦ They emit signals

⑧ An image is formed

Main Components of an MRI System

① **Magnet**

Types of magnets:

- Superconducting
- Permanent
- Resistive

② **Gradient Coils**

Function:

Spatial localization of the signal

③ **RF Coil (Radiofrequency Coil)**

Function:

Transmits and receives RF signals

④ **Computer System**

Function:

☞ Processes signals and reconstructs the image

How the Image is Formed in MRI

① Hydrogen nuclei emit signals
② The RF coil receives the signals
③ The computer processes the signals
④ The image is reconstructed

Physical Concept of MRI

1-Hydrogen Nucleus

The hydrogen nucleus consists of:

- One proton

The proton has **spin** (intrinsic angular momentum)

This spin produces:

A **magnetic moment**

That means:

The hydrogen nucleus behaves like a very small magnet

MRI Depends on Three Main Elements

1- A strong magnetic field

2- Radiofrequency (RF) waves

3- Hydrogen nuclei

What Happens Without a Magnetic Field?

Inside the human body:

- The nuclei are randomly distributed
- Their directions are different

👉 The net result = **zero magnetization**

What Happens to the Nuclei in a Magnetic Field?

When a strong magnetic field is applied:

Hydrogen nuclei align with the magnetic field in two ways:

① **Parallel to the field (Low energy state)**
② **Anti-parallel to the field (High energy state)**

However:

- The number of parallel nuclei is slightly greater

👉 Therefore, a **net magnetization** is produced

Motion of the Nuclei (Precession)

Hydrogen nuclei do not remain stationary

They rotate around the magnetic field

This motion is called:

Precession

It is similar to:

☞ The motion of a spinning top

Larmor Frequency

The nuclei rotate at a specific frequency called:

Larmor frequency

$\omega = \gamma B_0$

Meaning of the symbols:

- **ω** = angular frequency
- **γ** = gyromagnetic ratio
- $\mathbf{B_0}$ = magnetic field strength

As the magnetic field increases, the frequency increases

Example

In a **1.5 Tesla MRI scanner**:

☞ Hydrogen frequency ≈ **63 MHz**

4-Radiofrequency (RF Pulse)

Now we apply:

☞ RF waves at the **Larmor frequency**

What happens?

☞ **Resonance occurs**

What is Resonance?

Resonance means:

☞ Absorption of energy at a specific frequency

Example:

Like tuning a radio to a specific station

What Happens After Applying RF?

① The nuclei absorb energy
② They tilt away from the magnetic field
③ The RF pulse is turned off
④ The nuclei return to their original alignment
⑤ They emit a signal

This signal is measured

- It is received by the **RF coil**
- Then processed by the computer to form an image

3-When the Body is Placed in a Strong Magnetic Field

Inside the MRI scanner:

☞ A strong magnetic field called **B_0** is present

Typically:

- **1.5 Tesla**
- **3 Tesla**

Why Hydrogen Specifically?

What Happens Without a Magnetic Field?

- Nuclei are randomly oriented

- They move in all directions

Net magnetization = **zero**

What Happens in a Magnetic Field (B_0)?

The nuclei begin to align with the field in two states:

① Parallel (Low Energy State)

1-Lower energy

2-More stable

② Anti-parallel (High Energy State)

1- Higher energy

2- Less stable

Why Are More Nuclei Parallel?

Because:

Physical systems prefer **lower energy states**

So:

More protons align **parallel** to the field

What Is Produced?

Since:

☞ Parallel > Anti-parallel

☞ A small difference appears

This produces:

☞ **Net Magnetization**

What is Net Magnetization?

☞ It is the **overall resultant of all magnetic moments**

Simple Example

Assume:

- 1000 protons
- 500 parallel
- 495 anti-parallel

☞ Difference = 5 protons

This small difference:

☞ Produces the MRI signal

Direction of Net Magnetization

☞ It aligns with the magnetic field

This is called:

☞ **Longitudinal Magnetization**

What Happens Next?

After that:

☞ An RF pulse is applied

Resulting in:

- Energy absorption
- Magnetization tilting
- Signal emission

Summary of the Physical Concept

✓ The proton has spin

✓ Spin produces a magnetic moment

✓ The magnetic field aligns the nuclei

✓ More nuclei are parallel

✓ Net magnetization is produced

✓ This is the basis of MRI

Why is Hydrogen Signal Strong?

✓ Hydrogen is abundant in the body

Example:

☞ The human body is approximately **70% water**

Since water contains hydrogen:

☞ The MRI signal is **very strong**



This is Magnetic Resonance Imaging (MRI)